

[House Robber]

n

maxAmount(n)

nums: $[1, 2, 3, 1]$

$$\begin{aligned} 1 + 3 &= 4 \\ 1 + 1 &= 2 \\ 2 + 1 &= 3 \\ 2 + &= 2 \end{aligned} \quad \left. \vphantom{\begin{aligned} 1 + 3 \\ 1 + 1 \\ 2 + 1 \\ 2 + \end{aligned}} \right\} = 4$$

maxAmount(1)

maxAmount(0)

house $\rightarrow [1, 7, 2, 3, 1, 9]$

Base case

$$\begin{aligned} 1 + 2 + 1 &\rightarrow 4 \\ 1 + 3 + 9 &\rightarrow 13 \\ 7 + 3 + 9 &\rightarrow 19 \end{aligned}$$

int maxAmount(n)

```
{
    if (n == 0)
        return house[0]
    if (n == 1)
        return max(house[0], house[1]);
}
```

$$\text{return } \max(\text{house}[n] + \text{maxAmount}(n-2), \text{maxAmount}(n-1))$$

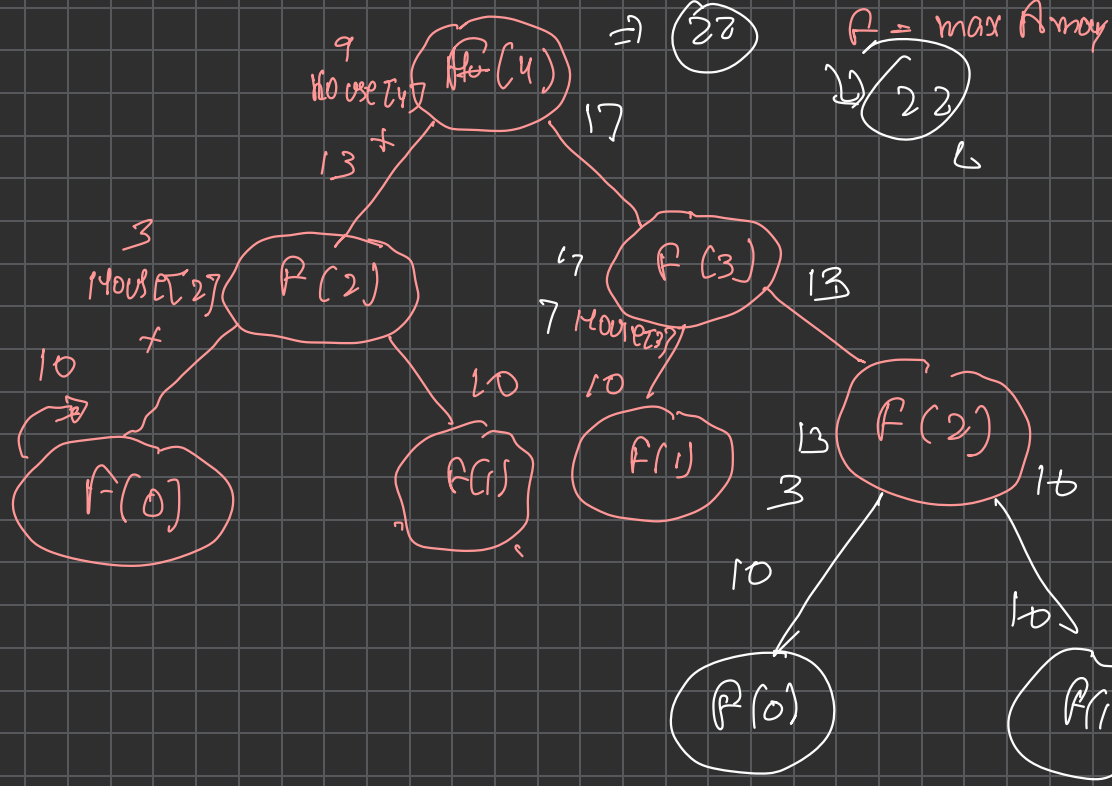
4

maxAmount(n)

house $\rightarrow$ [10, 2, 3, 7, 9]

$\Rightarrow$ (22)

$P = \text{max Array}$
 $\rightarrow$ (22)



$i=0$

$[10, 3, 4, 8, 9]$

$n=5$

int maxAmount(i, n)

8

$i == n-1$

$P(0, n)$

10

13

$P(2, n)$

$P(1, n)$

$P(4, n)$

$P(3, n)$

n ✓

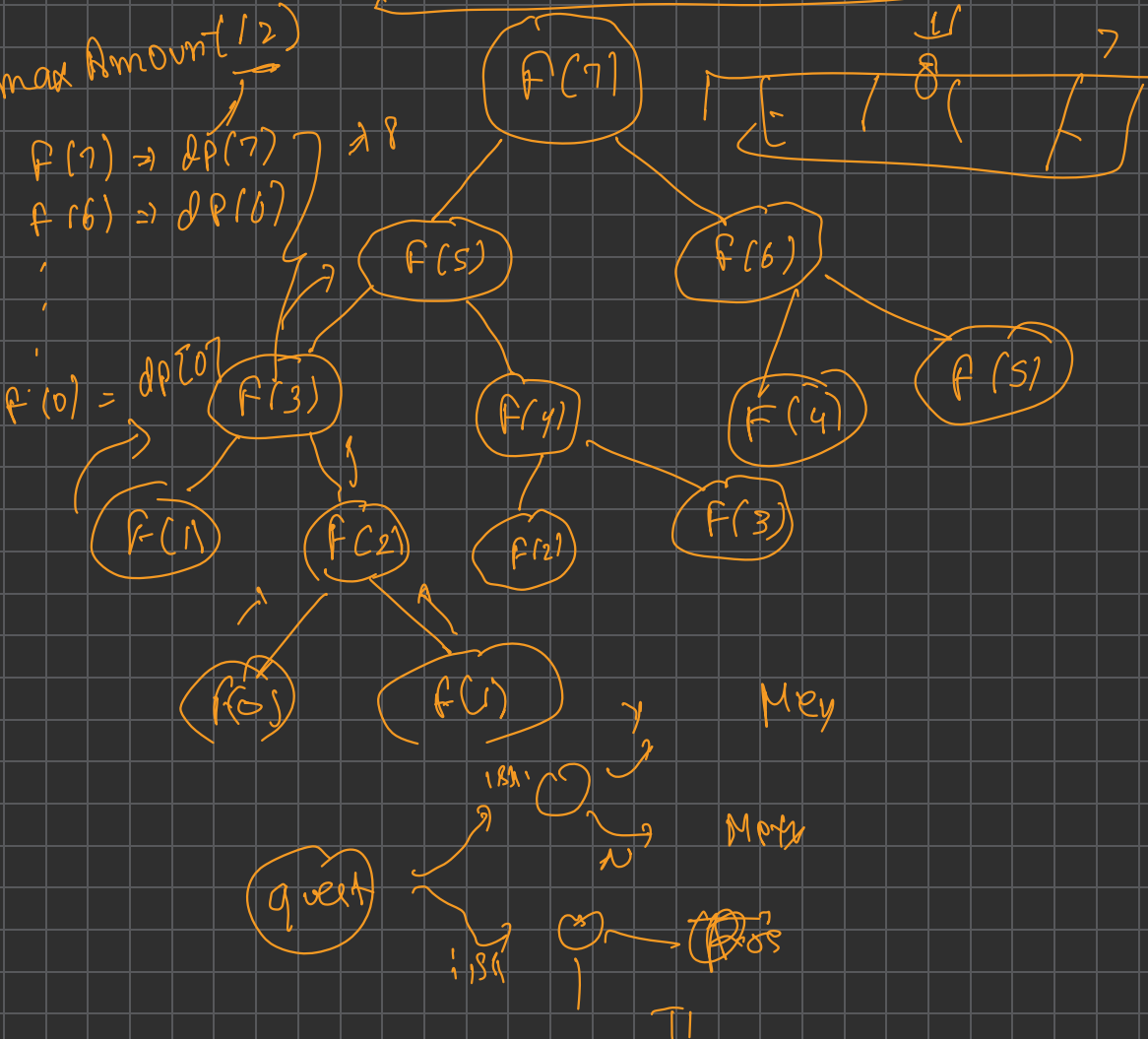
$n+1 \Rightarrow n+2$
 n

$dp \Rightarrow$

0 1 2 3 4 5 6 7
[10, 6, 8, 11, 13, 9, 8, 4]

max Amount(12)

$F(7) \Rightarrow dp(7) \Rightarrow 8$
 $F(6) \Rightarrow dp(6)$
:
 $F(0) = dp(0)$



May

Math

quest

TI

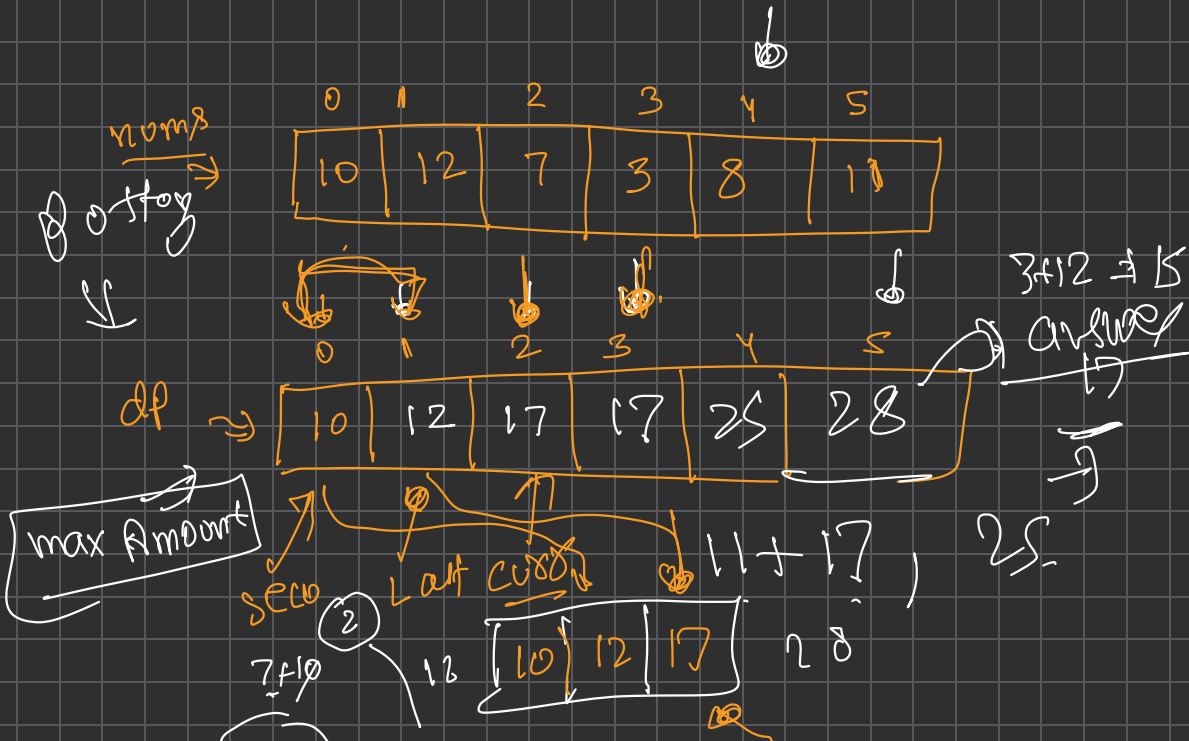
1000 1100 1200 1300

1400 1500 1600 1700

To

F

$O(n)$



for ($i = 2$; $i < n$; $i++$)

$$dp[i] = \max(nums[i] + dp[i-2], dp[i-1])$$

7 ↑

ext

for [i = 2; i < n; i++]
{
 dp[0] = dp[1]
 dp[1] = dp[2]
 dp[2] = max[nums[i] + dp[0], dp[1]]
}

→ space →

[int second_last, last, curr]

max [first, last+1]

max $\rightarrow$ 25 $\rightarrow$ $2+8+7$ $\rightarrow$ 17

$0 = n - 2$ $[10, 2, 3, 8, 9, ?]$

$1 = n - 1$ $[2, 3, 8, 9, 7, 4]$

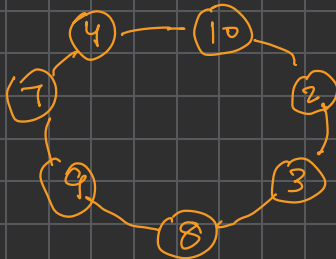
$0 \quad 1 \quad 2 \quad 3 \quad 4 \quad 5 \quad 6 \quad 7$

$[10, 2, 3, 8, 9, 7, 4]$

$3+9+4$

11

16



First	Last
✓	✗
✗	✓
✗	✗
✗	✓
✓	✗

$$10 + 3 + 9 = 22$$

$$10 + 8 + 7 = 25$$

$$2 + 8 + 7 = 17$$

$[10 \quad 40 \quad 8]$

(40)

$[8, 40, 6, 70, 3]$

x

(110)

